

EC400 Probability and Statistical Inference
Sample Exam 2: Solutions and Marking Guide
September 2026

If you need an additional assumption in order to proceed, state it clearly before using it.

Question 1

Employees in one branch of a firm may choose whether to attend an optional coaching programme. Let $D = 1$ denote attendance, let $Y(d)$ be the performance score that would be observed under programme status d , and let $Y = DY(1) + (1 - D)Y(0)$. Assume all relevant expectations exist, and write $p = \Pr(D = 1)$, where $0 < p < 1$. Define $\beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$. **[25 marks]**

- (a) Define the individual treatment effect, the average treatment effect and the average treatment effect on the treated. Explain why only one potential outcome is observed for each employee. **[5 marks]**

Solution: The individual treatment effect is

$$\tau_i = Y_i(1) - Y_i(0).$$

The branch-population average treatment effect and average treatment effect on the treated are, respectively,

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)], \quad \text{ATET} = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

The observation rule is $Y = DY(1) + (1 - D)Y(0)$. Hence an attendee reveals $Y(1)$ and a non-attendee reveals $Y(0)$; the other potential outcome is counterfactual for that employee.

Marking guide: 1 mark for the individual effect; 1 for the ATE; 1 for the ATET; 2 for using the observation rule to explain the missing counterfactual.

- (b) Derive the decomposition of β into the ATET and a selection-bias term. Explain what each term represents. **[5 marks]**

Solution: The observation rule gives $\mathbb{E}[Y \mid D = 1] = \mathbb{E}[Y(1) \mid D = 1]$ and $\mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(0) \mid D = 0]$. Adding and subtracting $\mathbb{E}[Y(0) \mid D = 1]$ therefore gives

$$\begin{aligned} \beta &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \\ &= \text{ATET} + [\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]]. \end{aligned}$$

The ATET is the mean causal effect for employees who attend. The second term compares the attendees' and non-attendees' mean outcomes under the same untreated state. It captures pre-existing selection rather than a programme effect. Thus the observed mean difference generally combines causation and selection.

Marking guide: 1 mark for applying the observation rule; 2 for the correct decomposition; 1 for interpreting the ATET; 1 for interpreting selection bias.

- (c) Use the law of iterated expectations to prove that

$$\text{ATE} = p \text{ ATET} + (1 - p)\mathbb{E}[Y(1) - Y(0) \mid D = 0].$$

Explain separately why the ATE may differ from the ATET and why β may differ from the ATET. **[5 marks]**

Solution : Conditioning on D gives

$$\begin{aligned} \text{ATE} &= \mathbb{E}[\mathbb{E}[Y(1) - Y(0) \mid D]] \\ &= p \mathbb{E}[Y(1) - Y(0) \mid D = 1] \\ &\quad + (1 - p) \mathbb{E}[Y(1) - Y(0) \mid D = 0] \\ &= p \text{ ATET} + (1 - p) \mathbb{E}[Y(1) - Y(0) \mid D = 0]. \end{aligned}$$

The ATE differs from the ATET when treatment effects differ systematically between attendees and non-attendees. By contrast, the decomposition in part (b) shows that β differs from the ATET when the two groups have different mean untreated potential outcomes. Differences in average treatment effects and selection bias are separate issues.

Marking guide: 3 marks for the iterated-expectations proof; 1 for the treatment-effect explanation; 1 for the selection-bias explanation.

- (d) Suppose instead that programme status were randomly assigned within this branch. State the resulting potential-outcome independence condition and its implications for β , the ATET and the ATE. Would this alone identify the ATE for every branch of the firm? **[5 marks]**

Solution : Random assignment within the branch gives

$$D \perp\!\!\!\perp (Y(0), Y(1))$$

within that study population. It makes both potential-outcome means equal across assignment groups, so the selection-bias term is zero and

$$\beta = \text{ATET} = \text{ATE}$$

for the randomized branch population. It does not by itself identify a firm-wide ATE: employees in this branch need not represent employees elsewhere, and treatment effects may differ across branches. Random assignment identifies the effect within the study branch. Applying it to other branches requires suitable random sampling or additional assumptions.

Marking guide: 1 mark for the independence condition; 1 for the zero selection-bias implication; 1 for $\beta = \text{ATET} = \text{ATE}$ within the branch; 2 for the distinction between identifying the effect in this branch and applying it to other branches.

- (e) In an observational sample, let n_1 and n_0 be the numbers of attendees and non-attendees, and let $\bar{Y}_1$ and $\bar{Y}_0$ be their sample means. State a natural estimator of β . Distinguish an estimator, an estimate and a parameter. Under random sampling within each group, show why increasing both group sizes improves estimation of β but cannot by itself remove selection bias. **[5 marks]**

Solution : A natural estimator is

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0,$$

The estimator is this random rule before the sample is observed; an estimate is its realized numerical value; and the parameter is the fixed population comparison β . Under random sampling within groups, the law of large numbers gives

$$\bar{Y}_1 \xrightarrow{p} \mathbb{E}[Y \mid D = 1], \quad \bar{Y}_0 \xrightarrow{p} \mathbb{E}[Y \mid D = 0],$$

so $\hat{\beta} \xrightarrow{p} \beta$. Larger samples therefore improve estimation of β . But if self-selection makes the untreated potential-outcome means differ, part (b) shows that β still differs from the causal target. Consistent estimation of the observed difference cannot repair a failure of causal identification.

Marking guide: 1 mark for the estimator; 1 for distinguishing the three objects; 1 for the LLN argument; 2 for separating estimation from identification.

Question 2

The joint density of continuous random variables (X, Y) is

$$f_{X,Y}(x, y) = x + y, \quad 0 < x < 1, \quad 0 < y < 1,$$

and is zero otherwise.

[25 marks]

- (a) Derive the two marginal densities and the conditional density of X given $Y = y$. State all relevant supports.

[6 marks]

Solution: For $0 < x < 1$ and $0 < y < 1$,

$$f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2}, \quad f_Y(y) = \int_0^1 (x + y) dx = y + \frac{1}{2}.$$

Both marginals are zero outside $(0, 1)$. Thus, for $0 < y < 1$,

$$f_{X|Y}(x | y) = \frac{x + y}{y + 1/2}, \quad 0 < x < 1,$$

and it is zero otherwise. The denominator is positive throughout the stated conditioning support.

Marking guide: 2 marks for the two marginal densities; 2 for the conditional-density formula; 2 for the correct marginal, conditioning and conditional supports.

- (b) Calculate $\mathbb{E}[X | Y = y]$. Then calculate $\mathbb{E}[X]$ directly from f_X and confirm that averaging the conditional mean recovers the same result. **[6 marks]**

Solution: For $0 < y < 1$,

$$\begin{aligned}\mathbb{E}[X | Y = y] &= \frac{\int_0^1 x(x+y) dx}{y + 1/2} \\ &= \frac{1/3 + y/2}{y + 1/2}.\end{aligned}$$

The marginal calculation gives

$$\mathbb{E}[X] = \int_0^1 x \left(x + \frac{1}{2} \right) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}.$$

Iterating gives exactly the same answer:

$$\begin{aligned}\mathbb{E}[\mathbb{E}(X | Y)] &= \int_0^1 \frac{1/3 + y/2}{y + 1/2} \left(y + \frac{1}{2} \right) dy \\ &= \int_0^1 \left(\frac{1}{3} + \frac{y}{2} \right) dy = \frac{7}{12}.\end{aligned}$$

Marking guide: 2 marks for the conditional-expectation calculation; 2 for the direct marginal mean; 2 for an explicit and correct iterated-expectations check.

- (c) Calculate $\mathbb{E}[XY]$ and $\text{Cov}(X, Y)$. Are X and Y independent? **[7 marks]**

Solution: We have

$$\begin{aligned}\mathbb{E}[XY] &= \int_0^1 \int_0^1 xy(x+y) dx dy \\ &= \left(\int_0^1 x^2 dx \right) \left(\int_0^1 y dy \right) + \left(\int_0^1 x dx \right) \left(\int_0^1 y^2 dy \right) \\ &= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.\end{aligned}$$

Symmetry gives $\mathbb{E}[Y] = 7/12$, so

$$\text{Cov}(X, Y) = \frac{1}{3} - \left(\frac{7}{12} \right)^2 = -\frac{1}{144}.$$

The variables are not independent. In particular, their non-zero covariance rules out independence; equivalently, the conditional density found above depends on y .

Marking guide: 3 marks for $\mathbb{E}[XY]$; 2 for the covariance; 2 for the correct independence conclusion with a valid justification.

(d) Let $W = X + Y$. Calculate $\mathbb{E}[W]$ and $\text{Var}(W)$.

[6 marks]

Solution: First,

$$\mathbb{E}[X^2] = \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}.$$

Hence

$$\text{Var}(X) = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{11}{144}.$$

By symmetry, $\text{Var}(Y) = 11/144$. Therefore

$$\mathbb{E}[W] = \frac{7}{12} + \frac{7}{12} = \frac{7}{6},$$

$$\text{Var}(W) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) = \frac{11 + 11 - 2}{144} = \frac{5}{36}.$$

Marking guide: 1 mark for $\mathbb{E}[X^2]$; 2 for $\text{Var}(X)$ and symmetry; 1 for $\mathbb{E}[W]$; 2 for $\text{Var}(W)$.

Question 3

Let $U \sim U(0, 1)$, let $\lambda > 0$, and define

$$T = -\frac{1}{\lambda} \log(1 - U).$$

This transformation is strictly increasing. Let $T_1, T_2, \dots$ be independent copies of T , and let $S_m = \sum_{j=1}^m T_j$, where m is a positive integer. **[25 marks]**

Reference formulas. You may use the following, with $b, \nu > 0$ and positive integer a :

- If $W \sim \text{Exp}(b)$, then $\mathbb{E}[W] = 1/b$ and $\text{Var}(W) = 1/b^2$.
- A Gamma(a, b) variable (shape a , rate b) has density

$$f_G(s) = \frac{b^a}{(a-1)!} s^{a-1} e^{-bs}, \quad s > 0,$$

and zero otherwise.

- If $N \sim \text{Poisson}(\nu)$, then

$$\Pr(N = j) = e^{-\nu} \frac{\nu^j}{j!}, \quad j = 0, 1, \dots, \quad \mathbb{E}[N] = \text{Var}(N) = \nu.$$

- (a) State the support of T and find the inverse transformation. **[5 marks]**

Solution: The transformation maps $(0, 1)$ onto $(0, \infty)$. Solving $t = -\lambda^{-1} \log(1 - u)$ gives

$$u = 1 - e^{-\lambda t}, \quad t > 0.$$

Marking guide: 2 marks for the support; 3 for the inverse.

- (b) Use the CDF method to derive the CDF and density of T over the whole real line. Name the distribution and give its parameter. **[5 marks]**

Solution: For $t > 0$, monotonicity and the uniform CDF give

$$\begin{aligned} F_T(t) &= \Pr[-\lambda^{-1} \log(1 - U) \leq t] \\ &= \Pr(U \leq 1 - e^{-\lambda t}) = 1 - e^{-\lambda t}. \end{aligned}$$

Thus

$$F_T(t) = \begin{cases} 0, & t \leq 0, \\ 1 - e^{-\lambda t}, & t > 0, \end{cases} \quad f_T(t) = \begin{cases} \lambda e^{-\lambda t}, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

Therefore T is exponential with rate λ .

Marking guide: 2 marks for the complete CDF derivation; 1 for the density and its support; 2 for the correctly parameterised distribution.

- (c) Derive the tail probability $\Pr(T > t)$ and median of T . Give its expectation and variance, and explain how increasing λ changes the distribution. **[5 marks]**

Solution: For $t > 0$, $\Pr(T > t) = 1 - F_T(t) = e^{-\lambda t}$; for $t \leq 0$ it equals one. The median solves $F_T(q_{1/2}) = 1/2$, so $q_{1/2} = (\log 2)/\lambda$. The supplied exponential moment formulas give

$$\mathbb{E}[T] = \frac{1}{\lambda}, \quad \text{Var}(T) = \frac{1}{\lambda^2}.$$

Increasing λ lowers the mean, median and standard deviation, and reduces $\Pr(T > t)$ for every fixed $t > 0$: waiting times become shorter.

Marking guide: 1 mark for the tail probability; 2 for the median; 1 for the expectation and variance; 1 for the comparative interpretation.

- (d) State the distribution of S_m , including its parameterisation. Write its density and give its expectation and variance. **[5 marks]**

Solution: A sum of m independent exponential waiting times with common rate λ has a $\text{Gamma}(m, \lambda)$ distribution: shape m , rate λ . Substituting into the supplied density gives

$$f_{S_m}(s) = \frac{\lambda^m}{(m-1)!} s^{m-1} e^{-\lambda s}, \quad s > 0.$$

The density is zero otherwise. By linearity of expectation and independence of the T_j , its moments are

$$\mathbb{E}[S_m] = \sum_{j=1}^m \mathbb{E}[T_j] = \frac{m}{\lambda}, \quad \text{Var}(S_m) = \sum_{j=1}^m \text{Var}(T_j) = \frac{m}{\lambda^2}.$$

Marking guide: 1 mark for the shape–rate distribution; 2 for the density; 1 for the expectation; 1 for the variance.

- (e) Interpret the T_j as successive waiting times between arrivals and, for $t > 0$, let $N(t)$ be the number of arrivals by time t . State the distribution of $N(t)$, express $\Pr(S_m \leq t)$ in terms of $N(t)$, and derive its probability. Also give $\mathbb{E}[N(t)]$ and $\text{Var}(N(t))$. **[5 marks]**

Solution: The arrival rate is λ per unit of time, so

$$N(t) \sim \text{Poisson}(\lambda t).$$

The m th arrival occurs by time t exactly when at least m arrivals have occurred, so $S_m \leq t$ exactly when $N(t) \geq m$. Therefore

$$\Pr(S_m \leq t) = 1 - \Pr[N(t) \leq m-1] = 1 - e^{-\lambda t} \sum_{k=0}^{m-1} \frac{(\lambda t)^k}{k!}.$$

Here we used the supplied Poisson PMF with $\nu = \lambda t$. The supplied moment formulas likewise give $\mathbb{E}[N(t)] = \text{Var}(N(t)) = \lambda t$.

Marking guide: 1 mark for the Poisson distribution; 1 for the event equivalence; 2 for the probability; 1 for the moments.

Question 4

For four observations, let

$$z = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 6 \end{pmatrix}.$$

An intercept is included in the linear regression of y on z . You may use the familiar OLS formula $\hat{\beta} = (X'X)^{-1}X'y$; no derivation of this formula is required. **[25 marks]**

(a) Write the design matrix X and calculate $X'X$. **[4 marks]**

Solution: With the intercept in the first column,

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

Direct multiplication gives

$$X'X = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

Marking guide: 2 marks for X ; 2 for the complete and correct $X'X$.

(b) Calculate $X'y$ and $(X'X)^{-1}$, and hence obtain the OLS intercept and slope. **[5 marks]**

Solution: The required products are

$$X'y = \begin{pmatrix} 12 \\ 25 \end{pmatrix}, \quad (X'X)^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

Therefore

$$\hat{\beta} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix} \begin{pmatrix} 12 \\ 25 \end{pmatrix} = \begin{pmatrix} 9/10 \\ 7/5 \end{pmatrix}.$$

The fitted intercept is 0.9 and the fitted slope is 1.4.

Marking guide: 1 mark for $X'y$; 2 for the inverse; 1 for the matrix multiplication; 1 for both coefficient values.

(c) Calculate the fitted values and residuals. Verify $X'\hat{u} = 0$ and explain why a fitted residual is not the same object as the unobserved population error ε_i . **[5 marks]**

Solution: The fitted values and residuals are

$$\hat{y} = X\hat{\beta} = \begin{pmatrix} 0.9 \\ 2.3 \\ 3.7 \\ 5.1 \end{pmatrix}, \quad \hat{u} = y - \hat{y} = \begin{pmatrix} 0.1 \\ 0.7 \\ -1.7 \\ 0.9 \end{pmatrix}.$$

Their sum is zero, and their z -weighted sum is

$$0(0.1) + 1(0.7) + 2(-1.7) + 3(0.9) = 0.$$

Thus $X'\hat{u} = (0, 0)'$. A residual is an observable, fitted-sample gap $y_i - x_i'\hat{\beta}$. The population error is the unobserved DGP component $\varepsilon_i = y_i - x_i'\beta$ defined using the unknown population coefficient.

Marking guide: 1 mark for the fitted values; 1 for the residuals; 1 for both checks that both entries of $X'\hat{u}$ are zero; 2 for distinguishing residuals from population errors.

- (d) Suppose the conditional covariance matrix of the population errors is

$$\Omega = \text{Var}(\varepsilon | X) = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Describe the variance and dependence pattern. Calculate $X'\Omega X$ and use

$$\text{Var}(\hat{\beta} | X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$$

to obtain the exact conditional covariance matrix of $\hat{\beta}$ and the two corresponding standard errors. In a large sample with this covariance pattern and suitable technical conditions, which robust standard-error method is appropriate, and why does it leave $\hat{\beta}$ unchanged? **[7 marks]**

Solution: The diagonal entries show conditional error variances 2, 5, 1, 3, so the errors are heteroscedastic. Every off-diagonal covariance is zero, so the matrix encodes no cross-observation correlation. Direct multiplication gives

$$X'\Omega X = \begin{pmatrix} 11 & 16 \\ 16 & 36 \end{pmatrix}.$$

Using the inverse from part (b),

$$\begin{aligned} \text{Var}(\hat{\beta} | X) &= \frac{1}{400} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix} \begin{pmatrix} 11 & 16 \\ 16 & 36 \end{pmatrix} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix} \\ &= \frac{1}{100} \begin{pmatrix} 191 & -79 \\ -79 & 51 \end{pmatrix}. \end{aligned}$$

Thus the intercept and slope standard errors are $\sqrt{191}/10$ and $\sqrt{51}/10$, respectively. This exact calculation uses the supplied population covariance Ω . In a large sample with this heteroscedastic, cross-observation-uncorrelated pattern and suitable technical conditions, White standard errors are appropriate. They change the estimated sampling covariance of $\hat{\beta}$, not the OLS objective or coefficient formula. Hence the fitted coefficient remains $(0.9, 1.4)'$ while its reported uncertainty changes.

Marking guide: 1 mark for the variance and dependence pattern; 2 for $X'\Omega X$; 2 for the covariance matrix calculation; 1 for the two standard errors and White method; 1 for explaining why the OLS coefficients do not change.

- (e) Suppose z_i records self-selected training hours. Interpret the estimated slope and explain why it is not automatically a causal effect. What does $\mathbb{E}[\varepsilon_i \mid z_i] = 0$ imply, and what kind of design would strengthen a causal interpretation? **[4 marks]**

Solution: The slope 1.4 is the fitted increase in the outcome associated with one additional training hour in this sample. The condition

$$\mathbb{E}[\varepsilon_i \mid z_i] = 0$$

makes the population line $\beta_0 + \beta_1 z_i$ the conditional mean of y_i given z_i and supports the usual regression interpretation. Self-selection can nevertheless make training hours related to motivation, prior skill or other determinants of outcomes, so the fitted association is not automatically causal. Random assignment of training intensity, or another valid design that makes training unrelated to other causes of the outcome, would strengthen the causal interpretation.

Marking guide: 1 mark for the sample slope and zero-conditional-mean interpretations; 2 for explaining the self-selection problem; 1 for naming a valid design or identification requirement.

Question 5

For an estimator $\hat{\beta}_n$, suppose that, for every sample size n ,

$$\hat{\beta}_n \sim N\left(\beta, \frac{\sigma^2}{n}\right),$$

where $\sigma > 0$ is known. Before observing the data, a researcher chooses to test $H_0 : \beta \leq 0$ against $H_1 : \beta > 0$ at significance level α . Write $z_{1-\alpha}$ for the corresponding standard-Normal quantile and let

$$Z_n = \frac{\sqrt{n}\hat{\beta}_n}{\sigma}.$$

The null includes zero and negative effects. To control the chance of rejecting a true null, we must consider every value $\beta \leq 0$. The largest of these rejection probabilities is called the *size* of the test.

[25 marks]

- (a) Consider the rule that rejects when $Z_n > z_{1-\alpha}$. Find its rejection probability for a value $\beta \leq 0$. Show that this probability is largest at $\beta = 0$, where it equals α , so the test has size α . Explain why the direction of the test must be chosen before the estimate is observed. **[5 marks]**

Solution: Under any value of β ,

$$Z_n \sim N\left(\frac{\sqrt{n}\beta}{\sigma}, 1\right).$$

Hence the rejection probability is

$$\Pr_{\beta}(Z_n > z_{1-\alpha}) = 1 - \Phi\left(z_{1-\alpha} - \frac{\sqrt{n}\beta}{\sigma}\right).$$

This probability is increasing in β . Over the null values $\beta \leq 0$, it is therefore largest at $\beta = 0$, where it equals $1 - \Phi(z_{1-\alpha}) = \alpha$. Choosing the favourable tail after seeing the sign of the estimate would enlarge the rejection region and invalidate the claimed one-sided size.

Marking guide: 2 marks for the distribution and rejection probability; 1 for monotonicity; 1 for the maximum at $\beta = 0$ and the size; 1 for the pre-specification explanation.

- (b) Given an observed value z_{obs} , use the null value $\beta = 0$ from part (a) to obtain the right-tail p -value. Show that comparing it with α gives the same decision as comparing z_{obs} with the critical value. Give a correct frequentist interpretation of the p -value. **[5 marks]**

Solution: The tail probability under $\beta = 0$ is

$$p = 1 - \Phi(z_{\text{obs}}).$$

Because Φ is strictly increasing,

$$p < \alpha \iff \Phi(z_{\text{obs}}) > 1 - \alpha \iff z_{\text{obs}} > z_{1-\alpha}.$$

Thus the rules agree. The p -value is the probability, calculated under the null value $\beta = 0$, of obtaining a statistic at least as favourable to the alternative as the one observed. It is not the probability that the null is true.

Marking guide: 1 mark for the tail probability; 2 for the equivalence proof; 2 for the interpretation and the posterior-probability warning.

- (c) Consider the lower one-sided confidence set

$$C_{1-\alpha} = \left[\hat{\beta}_n - z_{1-\alpha} \frac{\sigma}{\sqrt{n}}, \infty \right).$$

Show that $\Pr_{\beta}(\beta \in C_{1-\alpha}) = 1 - \alpha$: this is the interval's repeated-sampling coverage. Show also that the interval excludes zero exactly when the test rejects, and explain the coverage statement in words. **[5 marks]**

Solution: For any fixed β , the supplied Normal distribution gives

$$\begin{aligned} \Pr_{\beta}(\beta \in C_{1-\alpha}) &= \Pr_{\beta} \left(\hat{\beta}_n - z_{1-\alpha} \frac{\sigma}{\sqrt{n}} \leq \beta \right) \\ &= \Pr_{\beta} \left(\frac{\sqrt{n}(\hat{\beta}_n - \beta)}{\sigma} \leq z_{1-\alpha} \right) \\ &= \Phi(z_{1-\alpha}) = 1 - \alpha. \end{aligned}$$

The set excludes zero exactly when

$$\hat{\beta}_n - z_{1-\alpha} \frac{\sigma}{\sqrt{n}} > 0 \iff Z_n > z_{1-\alpha},$$

which is exactly the rejection condition. Across repeated samples, the procedure has coverage $1 - \alpha$ for the fixed parameter. This is a statement about the long-run performance of the procedure, not a posterior probability assigned to the realized set.

Marking guide: 2 marks for the coverage calculation; 1 for showing the CI and test give the same decision; 2 for the repeated-sampling interpretation.

- (d) For a fixed alternative $\beta > 0$, derive the power of the test. Show that power increases with β and with n , and decreases with σ . What happens to power as $n \rightarrow \infty$? **[5 marks]**

Solution: Under the fixed alternative, the power is

$$\pi_n(\beta) = 1 - \Phi \left(z_{1-\alpha} - \frac{\sqrt{n}\beta}{\sigma} \right).$$

Because Φ is increasing, power rises when the standardized signal $\sqrt{n}\beta/\sigma$ rises. It is therefore increasing in β and n and decreasing in σ . For fixed $\beta > 0$, the standardized signal tends to infinity as $n \rightarrow \infty$, so the probability of rejecting this false null tends to one.

Marking guide: 2 marks for the power function; 2 for the three comparative-static statements; 1 for the limiting-power conclusion.

- (e) Suppose we carry out M tests. A *false rejection* means rejecting a null hypothesis that is true. We want the probability of *at least one* false rejection across all M tests to be at most α . This is often called controlling the *family-wise error probability*.

The Bonferroni rule tests each null at level α/M . Use the union bound to show that the probability of at least one false rejection is at most α , even without independence.

For an exact calculation, now assume the tests are independent and all nulls are true, with each test having actual rejection probability α/M (as at $\beta = 0$ in part (a)). First find the probability of no false rejections, then the probability of at least one. Compare the latter with α .

What is the cost of the smaller per-test level in terms of power?

[5 marks]

Solution: Let A_j be the event of a false rejection in test j . At least one false rejection occurs when any of the events A_j occurs, so the event of interest is $\bigcup_{j=1}^M A_j$. The union bound gives, without any independence assumption,

$$\Pr\left(\bigcup_{j=1}^M A_j\right) \leq \sum_{j=1}^M \Pr(A_j) \leq \sum_{j=1}^M \frac{\alpha}{M} = \alpha.$$

With independent tests and actual rejection probability α/M for each,

$$\Pr(\text{no false rejections}) = \prod_{j=1}^M (1 - \alpha/M) = (1 - \alpha/M)^M.$$

Thus the exact family-wise error probability is $1 - (1 - \alpha/M)^M \leq \alpha$, with the inequality following from the union bound. If actual null rejection probabilities are smaller than α/M , this expression is an upper bound rather than an equality. Bonferroni is valid under arbitrary dependence, but using a smaller critical level for every test reduces power and can be conservative, especially when the tests are strongly dependent.

Marking guide: 2 marks for the union bound; 2 for the independent exact probability under the stated assumptions and its comparison with α ; 1 for the power cost.

Question 6

Let u_t , for $t \in \mathbb{Z}$, be independent and identically distributed with $\mathbb{E}[u_t] = 0$ and $\text{Var}(u_t) = \sigma^2$, where $0 < \sigma^2 < \infty$. Consider the time series

$$X_t = \mu + u_t + u_{t-1}.$$

[25 marks]

- (a) Calculate $\mathbb{E}[X_t]$, $\text{Var}(X_t)$ and $\text{Cov}(X_t, X_{t-h})$ for $h = 1$ and for $|h| \geq 2$.

[5 marks]

Solution: Linearity gives $\mathbb{E}[X_t] = \mu$. Independence of u_t and u_{t-1} gives

$$\text{Var}(X_t) = \text{Var}(u_t + u_{t-1}) = 2\sigma^2.$$

Adjacent observations share exactly the shock u_{t-1} , so

$$\text{Cov}(X_t, X_{t-1}) = \sigma^2.$$

For $|h| \geq 2$, the two observations contain disjoint independent shocks and their covariance is zero.

Marking guide: 1 mark for the mean; 1 for the variance; 3 for the full covariance pattern.

- (b) State the definition of weak stationarity in terms of first and second moments, and verify that (X_t) is weakly stationary.

[5 marks]

Solution: Weak stationarity requires a time-invariant mean, a time-invariant finite variance, and covariances that depend on two dates only through their lag. Here

$$\mathbb{E}[X_t] = \mu, \quad \text{Var}(X_t) = 2\sigma^2,$$

for every t , and the covariance at lag h is

$$\gamma_h = \begin{cases} 2\sigma^2, & h = 0, \\ \sigma^2, & |h| = 1, \\ 0, & |h| \geq 2. \end{cases}$$

It depends only on h , so all weak-stationarity requirements hold.

Marking guide: 1 mark for the constant-mean requirement; 1 for constant finite variance; 1 for the lag-only covariance requirement; 2 for verifying all three with the derived moments.

- (c) For $\bar{X}_n = n^{-1} \sum_{t=1}^n X_t$, derive $\text{Var}(\bar{X}_n)$ and show that $\bar{X}_n \xrightarrow{p} \mu$. You may use the following probability bound: for any finite-variance random variable Z and any $a > 0$,

$$\Pr(|Z - \mathbb{E}[Z]| > a) \leq \frac{\text{Var}(Z)}{a^2}.$$

[5 marks]

Solution: Adding the variances and covariances of the observations gives

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \frac{1}{n^2} \left[n\gamma_0 + 2 \sum_{h=1}^{n-1} (n-h)\gamma_h \right] \\ &= \frac{1}{n^2} [2n\sigma^2 + 2(n-1)\sigma^2] \\ &= \frac{(4n-2)\sigma^2}{n^2} \rightarrow 0. \end{aligned}$$

Equivalently, collect the independent shocks:

$$n(\bar{X}_n - \mu) = u_0 + 2 \sum_{t=1}^{n-1} u_t + u_n.$$

The variance of the right-hand side is $[1 + 4(n-1) + 1]\sigma^2 = (4n-2)\sigma^2$, giving the same result. Since $\mathbb{E}[\bar{X}_n] = \mu$, the supplied probability bound gives

$$\Pr(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{(4n-2)\sigma^2}{n^2\varepsilon^2} \rightarrow 0$$

for every $\varepsilon > 0$. Hence $\bar{X}_n \xrightarrow{p} \mu$.

Marking guide: 3 marks for a correct variance calculation and its limit, using either covariances across time or the independent-shock representation; 2 for the consistency argument using the supplied probability bound.

- (d) A researcher incorrectly treats the observations as uncorrelated. Compare the resulting variance formula for $\bar{X}_n$ with the correct large- n variance. Does this understate or overstate uncertainty? Which standard-error method from Lecture 4 is designed to allow for this serial correlation?

[5 marks]

Solution: Ignoring all serial covariance terms gives

$$\text{Var}_{\text{uncorrelated}}(\bar{X}_n) = \frac{2\sigma^2}{n}.$$

The correct variance is

$$\frac{(4n-2)\sigma^2}{n^2} \sim \frac{4\sigma^2}{n}.$$

Thus ignoring serial covariance makes the reported variance too small. For a time-ordered sample with short-range serial correlation, HAC standard errors are the natural response.

Marking guide: 1 mark for the variance ignoring serial covariance; 2 for the correct asymptotic variance; 1 for the direction of the uncertainty error; 1 for selecting HAC inference.

(e) Now consider the time-series regression $y_t = x_t' \beta + \varepsilon_t$ and the algebraic identity

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{t=1}^n x_t x_t' \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n x_t \varepsilon_t \right).$$

State probability-limit conditions sufficient for OLS consistency, and explain how moving from cross-sectional to time-series data changes the argument. **[5 marks]**

Solution: Sufficient limits are

$$\frac{1}{n} \sum_{t=1}^n x_t x_t' \xrightarrow{p} Q, \quad Q \text{ nonsingular,}$$

and

$$\frac{1}{n} \sum_{t=1}^n x_t \varepsilon_t \xrightarrow{p} \mathbb{E}[x_t \varepsilon_t] = 0.$$

The latter is the exogeneity moment; nonsingularity supplies sufficient regressor variation. Substituting the limits into the displayed decomposition gives

$$\hat{\beta} - \beta \xrightarrow{p} Q^{-1} 0 = 0,$$

so OLS is consistent. The algebra does not change for time-series data. What changes are the assumptions used to justify the laws of large numbers: stationarity or other stability conditions and sufficiently weak dependence replace simple independent sampling.

Marking guide: 1 mark for the denominator limit and nonsingularity; 2 for the zero numerator limit and its exogeneity meaning; 1 for the consistency conclusion; 1 for explaining that the algebra is unchanged but the LLN assumptions differ.